

Distributed Vector and Graph Storage Architecture

Sharding and Partitioning

Scaling vector stores to billions of vectors requires consistent hashing and metadata-based partitioning across distributed nodes.

Replication & Consistency

Raft consensus protocol manages leader election and index state replication, guaranteeing linearizable reads and high availability during node failures.

Hybrid Storage Engines

Modern vector engines store high-dimensional indexes in RAM/NVMe SSDs while offloading raw document text to object storage like AWS S3 or MinIO.